

Original article

Concurrent treatment with gonadotropin-releasing hormone agonists for chemotherapy-induced ovarian damage in premenopausal women with breast cancer: A meta-analysis of randomized controlled trials

Bo Yang^a, Weiwei Shi^a, Junlan Yang^a, Hui Liu^a, Hong Zhao^a, Xiaoyan Li^b, Shunchang Jiao^{a,*}

^a Department of Oncology, General Hospital of Chinese PLA, Beijing 100853, China

^b Department of Lung Cancer, Affiliated Hospital of Academy of PLA Military Medical Sciences, Beijing 100071, China

ARTICLE INFO

Article history:

Received 12 October 2012

Received in revised form

30 November 2012

Accepted 17 December 2012

Keywords:

Meta-analysis

GnRH agonists

Ovarian damage

Breast cancer

Chemotherapy

ABSTRACT

Background: While chemotherapy significantly improves the prognosis of breast cancer patients, it also damages otherwise healthy organs, such as the ovaries. Gonadotropin-releasing hormone (GnRH) agonists may have a protective effect against chemotherapy-induced ovarian toxicity in premenopausal women being treated for breast cancer; however, studies of its clinical efficacy have reported conflicting results.

Objectives: This meta-analysis was designed to assess the collective data from previous studies of GnRH agonists administered concurrently with chemotherapy to prevent chemotherapy-induced ovarian toxicity in premenopausal women with breast cancer.

Methods: Electronic literature databases (Cochrane Library, Medline, and Embase) were searched for relevant randomized controlled trials (RCTs) published prior to April 2012. Only RCTs that compared GnRH agonists plus chemotherapy to chemotherapy alone for premenopausal women with breast cancer were selected. A random-effects model was used to calculate the risk ratios (RRs) for premature ovarian failure (POF) within one year after chemotherapy treatment and rates of resumed menses and spontaneous pregnancy during the follow-up period after cessation of treatment.

Results: Five RCTs composed of 528 patients (GnRH agonist combination, $n = 274$; chemotherapy alone, $n = 254$) were included in the meta-analysis. Significantly fewer women treated with GnRH agonist experienced post-chemotherapy POF, yielding a RR of 0.40 (vs. chemotherapy alone, 95% confidence interval [CI] 0.21–0.75). In contrast, both treatment groups experienced similar rates of resumed menses (RR = 1.31, 95% CI 0.93–1.85) and spontaneous pregnancy (RR = 0.96, 95% CI 0.20–4.56).

Conclusion: Concurrent administration of GnRH agonists during chemotherapy treatment of breast cancer in premenopausal women appears to protect against chemotherapy-related POF in the first year after treatment, but appears to have no effect on resumed menses or spontaneous pregnancy rates.

© 2012 Elsevier Ltd. All rights reserved.

Introduction

Breast cancer remains one of the most frequently diagnosed malignancies in women. Although the incidence trends of breast cancer decreased from 1998 through 2006, however, the probability of developing breast cancer before age 40 years is 1 for every 206 women.¹ Major advances in treatment of breast cancer have achieved significant benefit from adjuvant systemic chemotherapy in terms of prolonged disease-free and overall survival.^{2,3} The increased survival of patients with breast cancer has given rise to fertility problem related to cancer treatment. While the traditional

method of adjuvant chemotherapy to treat early stage breast cancer has proven to be even more effective at improving prognosis in this younger population of women,² it can damage the otherwise healthy ovaries. The manifestations of chemotherapy-induced ovarian toxicity include irregular menses or amenorrhea, which ultimately lead to premature ovarian failure (POF) and infertility. Therefore, preservation of the ovarian function following adjuvant chemotherapy has become a major concern of both patients and clinician-scientists.

Gonadotropin-releasing hormone (GnRH) is a key regulator of the hypothalamic–pituitary–gonadal reproductive axis, and studies of therapeutic administration of GnRH agonists have revealed that long-term exposure produces a reversible repression of the reproductive axis.⁴ Therefore, GnRH agonists were applied as

* Corresponding author. Tel.: +86 010 66939761; fax: +86 010 66939128.

E-mail address: jiaosc301@yahoo.cn (S. Jiao).

adjuvant endocrine therapy to silence ovarian activity during chemotherapy treatment, thereby protecting the ovary in premenopausal patients with breast cancer so that ovarian function may be restored upon therapy cessation. Many subsequent studies have demonstrated that GnRH agonists treatment is associated with a reduced rate of chemotherapy-associated ovarian toxicity.^{5–14} However, most of these studies were designed as retrospective and non-randomized trials, which precluded their ability to robustly assess ovarian and fertility preservation in the GnRH-treated patients.

Based on the previously published clinical evidence, two well-designed meta-analyses^{15,16} determined that GnRH agonists treatment during chemotherapy significantly benefited ovarian preservation in premenopausal women. Another meta-analysis¹⁷ provided evidence that the GnRH agonists treatment can preserve both ovarian function and fertility. In each of these meta-analyses, however, there was substantial heterogeneity in the types of disease, including hematologic diseases (lymphoma and leukemia), systemic lupus erythematosus, and breast cancer. Therefore, the strength of association between GnRH-related preservation of ovarian function with breast cancer remains controversial. The recent publication of two new RCTs^{13,14} that investigated the effects of GnRH agonists on ovarian preservation in breast cancer patients instigated our efforts to perform a focused meta-analysis using the larger set of accumulated clinical evidence.

Here, we present the first concentrated meta-analysis using the most comprehensive, up-to-date clinical data available in the public literature to quantitatively assess the efficacy of GnRH agonists to protect against chemotherapy-induced ovarian damage in premenopausal women with breast cancer.

Methods

Search strategy

The medical literature electronic databases of PubMed, Embase, and the Cochrane Library were searched for relevant randomized controlled trials (RCTs) published prior to April 2012. No language restrictions were applied. Potentially relevant RCTs were identified by various combinations of the following search terms: breast cancer, buserelin, triptorelin, goserelin, leuprolide, histrelin, nafarelin, gonadotropin-releasing hormone agonists, luteinizing hormone (LH)-releasing hormone agonists, chemotherapy, and ovarian failure. When multiple publications were identified for the same population, only the most recent publication was selected. In addition, the reference list of each selected relevant publication was manually searched to identify any additional eligible trials.

Trial selection

The following inclusion criteria were used to select RCTs for the meta-analysis: (1) patient group consisted of women who were premenopausal with histological diagnosis of breast cancer; (2) patients had been treated with GnRH agonists plus chemotherapy and compared to patients treated with chemotherapy alone; and (3) GnRH agonists interventions were administered concurrently with chemotherapy. For trials with more than two treatment arms, each valid pairwise comparison was considered separately. Trials were excluded based upon the following criteria: (1) patient group consisted of women with locally advanced or metastatic disease; (2) different chemotherapy regimens had been compared between the GnRH agonists and control groups; or (3) case–control design, reviews, letters, or case reports.

Types of outcome measures

The primary outcome measures were the proportion of POF within one year after completion of the chemotherapy treatment and the incidence of women with resumption of menses during the follow-up period after cessation of treatment. Amenorrhea means the absence of menstruation for 3 or more months in women with past menses. POF is defined by the individuals' investigators. Secondary outcome measures included the proportions of women with spontaneous pregnancy or adverse effects during the follow-up period after cessation of treatment. If more than one pregnancy was reported for a single patient, only the first pregnancy was included in order to avoid a possible unit-of-analysis error.

Data extraction and quality assessment

Two reviewers (Bo Yang and Weiwei Shi) independently extracted the data from each trial using a standardized form with predefined criteria that had been developed specifically for this review and which included the following items: (1) baseline demographics: author, year of publication, and country of study; (2) participants: sample size and age; (3) GnRH agonists intervention; (4) chemotherapy intervention; (5) duration of intervention; (6) outcome measures; and (7) adverse effects. Discrepancies between the extracted datasets were resolved by discussion.

The methodological quality of each study was assessed in accordance with the guidelines in the Cochrane reviewers' handbook.¹⁸ The following trial design features were assessed: (1) measured or unmeasured baseline characteristics due to the method by which trial participants had been selected or assigned; (2) care provided apart from the intervention being evaluated; (3) method by which outcomes were ascertained, diagnosed, and verified; and (4) withdrawal or exclusion of participants throughout the course of the trial. If all quality criteria were met, the trial was considered to have low risk of bias (score: A). If one or more of the quality criteria were only partially met, the trial was considered to have moderate risk of bias (score: B), and if one or more criteria not met, the trial was considered to have high risk of bias (score: C).

Statistical methods

Included trials were stratified according to the type of intervention and the subsequent comparisons of GnRH agonists plus chemotherapy vs. chemotherapy treatment. All analyses were performed with STATA statistical software (version 12.0; STATA Corp LP, College Station, TX, USA). Dichotomous data were used to calculate relative risk (RR) with a 95% confidence interval (CI). Homogeneity of RR across studies was assessed by using the Cochrane Q statistic (with $p < 0.10$ indicating statistically significant heterogeneity) and the I^2 statistic (with $<40\%$ indicating "heterogeneity might not be important" and $>75\%$ indicating "considerable heterogeneity", based on the suggestion of the Cochrane Handbook for Systematic Review of Interventions¹⁹). As there was substantial heterogeneity in the types of chemotherapy used between the different trials, a random effects model (Mantel–Haenszel heterogeneity) was used to calculate the pooled RR.

Potential publication bias was also assessed by both Begg's rank correlation test²⁰ and Egger's linear regression test,²¹ with $p < 0.10$ indicating statistical significance. Finally, sensitivity analysis was used to investigate the influence of a single study on the overall risk estimate, and was carried out by sequentially omitting one study at each turn with the metaninf algorithm in STATA. A p -value of <0.05 was considered statistically significant.

Results

Literature search

A total of 276 potentially relevant publications were identified by the initial electronic search. After reviewing the full-texts, only five studies^{6,10–14} met the inclusion criteria (Fig. 1).

Baseline characteristics and risk of bias for the individual trials

Characteristics of the five RCTs are listed in Table 1. The five trials were composed of 528 premenopausal breast cancer patients, including 274 who had received GnRH with chemotherapy and 254 who had received chemotherapy alone. GnRH treatment consisted of goserelin^{10–12} or triptorelin.^{13,14} Both of the GnRH agonists were administered once every four weeks throughout the chemotherapy treatment period. In most of the trials,^{12–14} the GnRH agonist treatment was initiated at least one week prior to the chemotherapy treatment.

All of the included trials were classified as having low or moderate risk of bias according to the methodological quality assessment.

GnRH agonist treatment is associated with a lower POF rate

Four of the RCTs^{10,12–14} evaluated POF rate within one year after the chemotherapy treatment had been completed. One trial¹⁰ assessed the rate within eight months after treatment, while the other three assessed the rate for 12 months. As shown in Fig. 2, although the heterogeneity among the four trials was insignificant ($I^2 = 48.7\%$, $p = 0.119$), we also used the random effect model because of the obvious clinical heterogeneity. Administration of GnRH agonists was associated with reduced POF rates (vs. chemotherapy alone; RR = 0.40, 95% CI 0.21–0.75).

GnRH agonist treatment is not associated with increased resumption of menses

All five of the RCTs evaluated the resumption of menses in relation to GnRH agonist treatment during the longest follow-up.

As shown in Fig. 3, the rates of menses resumption were not significantly different between the GnRH agonist-treated patients and the patients treated with chemotherapy alone. The RR of 1.31 (95% CI, 0.93–1.85) indicated no significant association with menses resumption. The heterogeneity among the five trials was significant ($I^2 = 92.2\%$, $p = 0.000$). However, there was some change of RR in a fixed-effect model (RR = 1.32; 95% CI, 1.16–1.50).

GnRH agonist treatment is not associated with increased spontaneous pregnancy rate

Three of the RCTs^{12–14} evaluated the spontaneous pregnancy rate in relation to GnRH agonist treatment during the longest follow-up. As shown in Fig. 4, although the heterogeneity among the three trials was not significant ($I^2 = 6.4\%$, $p = 0.344$), we also used the random effect model because of the obvious clinical heterogeneity. The rates of spontaneous pregnancy were not significantly different between the GnRH agonist-treated patients and the patients treated with chemotherapy alone. The RR of 0.96 (95% CI 0.20–4.56) indicated no significant association with spontaneous pregnancy.

GnRH agonist treatment is not associated with increased adverse effects

Two of the five RCTs^{12,13} reported adverse events experienced by patients in the treatment groups. Hot flashes, mood swings, urogenital symptoms, insomnia, headache, and sweating were reported. As shown in Fig. 5, the random effect model method was used because of the obvious clinical heterogeneity. The RR for adverse effects was 1.24 (95% CI 0.91–1.68) and the difference between the two treatment groups was not statistically significant ($p = 0.17$).

Sensitivity and publication bias analysis for the GnRH-related RR of POF

As shown in Fig. 6, the quantitative summary measure of RR (95% CI) for POF changed very little by sequential omission of individual trials. Moreover, the two trials with the most influence on the results,^{10,14} did not change the direction of the effect.

Neither the Begg's rank correlation test ($p = 1.00$) nor the Egger's linear regression test ($p = 0.925$) showed any evidence of publication bias for the trials reporting adjusted RR of POF rate.

Discussion

The major findings of the current meta-analysis provide prospective evidence that administration of GnRH agonists concurrently with chemotherapy can significantly decrease the POF rate in premenopausal women within one year after completion of the breast cancer treatment. Specifically, the POF rate of women who received GnRH agonists concurrently with chemotherapy is 60% lower than in their counterparts who received chemotherapy alone.

Premature (or primary) ovarian failure (also known as primary ovarian insufficiency) is a form of infertility that affects women under the age of 40. Clinically defined as primary or secondary amenorrhea lasting more than four consecutive months, POF can be diagnosed by increased serum level of follicle-stimulating hormone (FSH; to >40 IU/mL, confirmed by a second measurement one month later) coupled with decreased estrogen levels.^{22–25} However, the five RCTs included in the current meta-analysis used various definitions of POF to identify their study populations. Most of trials^{11,13,14} defined POF as no resumption of the menstrual cycle after cessation of chemotherapy, while one of the trials^{10,12,14} included measurements of FSH, LH, estradiol, or inhibin A and B

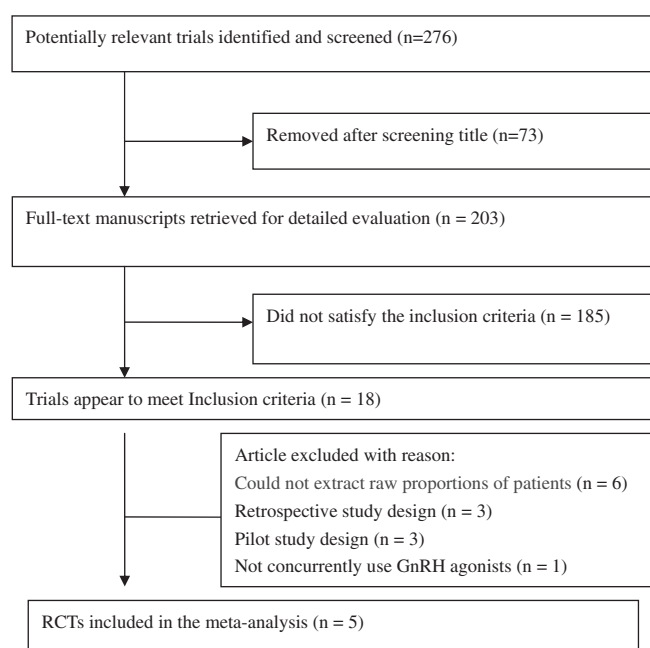


Fig. 1. Flow chart of trial selection process for meta-analysis.

Table 1

Summary of clinical studies included in the meta-analysis.

Study	Country	Design	Participants/Estrogen receptor status	Age: median (range) or mean $\pm$ SD (y)	Experimental group	Control	Follow-up (month)	Outcomes	Quality assessment
Badawy et al. 2009	Egypt	Randomized, control trial	Goserelin: 40 Control: 40 ER/PR NP	Goserelin: 30.00 $\pm$ 3.51 Control: 29.20 $\pm$ 2.93	Goserelin (3.6 mg) was administered every 4 weeks continuing for 6 months	FAC regimen (a combination of 5-fluorouracil 500 mg/m ² i.v., doxorubicin 500 mg/m ² i.v., and cyclophosphamide 500 mg/m ² i.v.) on day 1 of therapy to be repeated every 6–8 weeks for 6 cycles.	8	POF (Early cessation of menstruation, ovulation and increased serum FSH level). Incidence of spontaneous menstruation Incidence of spontaneous ovulation	B
Sverrisdottir et al. 2009	Sweden	Randomized multi-center trials	Goserelin alone (<i>n</i> = 29), Goserelin/Tamoxifen (<i>n</i> = 37), Tamoxifen alone (<i>n</i> = 29), no endocrine agent (<i>n</i> = 28). ER/PR NP	NP	Goserelin (3.6 mg)/28 days for 2 years	No GnRH analog. Up to 6 cycles of CMF regimen	12	Cessation of menstruation, resumption of menses within 1 year of the end of chemotherapy.	B
Gerber et al. 2011	German	Randomized, open-label, controlled multicenter trial	Goserelin 30 Control 30 ER/PR negative 30/30	Goserelin 35.0(26–44) Control 38.5(29–47)	Goserelin (3.6 mg) was administered at least 2 weeks before the first chemotherapy, continuing every 4 weeks until the end of the last cycle.	At least an anthracycline and cyclophosphamide with more than 500 mg/m ² per cycle and more than 2400 mg/m ² in total per regimen, administered every 3 weeks for six or eight cycles.	24	Normal ovarian function (two consecutive menstrual periods within 21–35 days of 5–8 months after last administration of goserelin). Time until recovery of regular menstruation; ovarian function (FSH, LH, E2), adverse events	B
Del Mastro et al. 2011	Italy	Parallel, randomized, open-label, trial	Triptorelin 148 Control 133 ER-positive, PR-positive, or both 109/117; ER and PR-negative 22/29 Unknown 2/2	Triptorelin 39 (24–45) Control 39 (25–45)	Triptorelin (3.75 mg) was administered at least 1 week before the start of chemotherapy, continuing every 4 weeks until the end of the last cycle.	Anthracycline, anthracycline + taxane, or CMF-based (100 mg/m ² of oral cyclophosphamide on days 1–14 or 600 mg/m ² of intravenous cyclophosphamide on days 1 and 8; 40 mg/m ² of methotrexate on days 1 and 8; and 600 mg/m ² of fluorouracil on days 1 and 8) chemotherapy.	12	POF (No resumption of menstrual activity and postmenopausal levels of FSH and estradiol 1 year after the last cycle of chemotherapy). Resumption of menses within 1 year of the end of chemotherapy.	A
Munster et al. 2012	USA	Randomized, double-blind, placebo-controlled trial	Triptorelin 27 Control 22 ER Positive 20/16; Negative 7/6 PR – positive 14/16; Negative 13/6	Triptorelin: 39(21–44) Control 38 (26–44)	Triptorelin (3.75 mg) was administered at least 1 weeks before the first chemotherapy every 28–30 days continuing every 4 weeks until the end of the last cycle.	Doxorubicin plus cyclophosphamide, four cycles of doxorubicin plus cyclophosphamide followed by four cycles of a taxane, or six cycles with fluorouracil plus epirubicin or doxorubicin plus cyclophosphamide for 4 cycles.	30	POF (as measured by the number of patients with interrupted or not restored menses during a follow-up period of at least 2 years after the last cycle of chemotherapy) Resumption of menses and serial monitoring of FSH and inhibin A and B levels.	A

Abbreviations: ER, estrogen receptor; PR, progesterone receptor; CMF, cyclophosphamide, methotrexate, and fluorouracil; NP, not report; FSH, follicle-stimulating hormone; LH, luteinizing hormone; POF, premature ovarian failure.

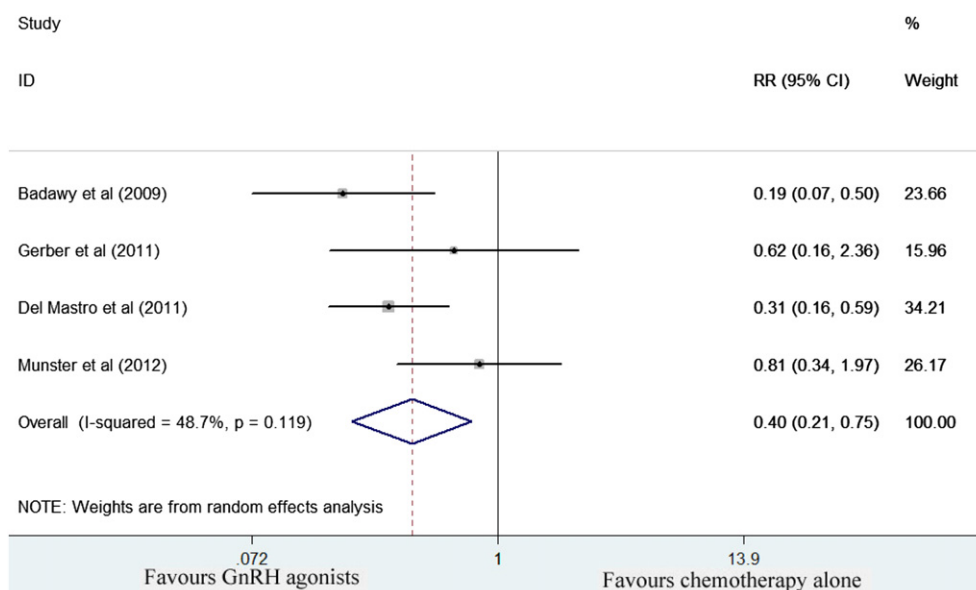


Fig. 2. Forest plots showing RR of POF rate of eligible studies comparing GnRH agonists plus chemotherapy with chemotherapy alone in a random effect model.

in their diagnoses. It is important to note that this difference between the trials may have impacted our meta-analysis results.

The incidence rates reported for permanent POF following systemic therapy for breast cancer have ranged from 33% to 76%.^{26,27} For example, Badawy et al.¹⁰ reported that POF in the control group (75%) may be related to the acute temporary effect of the chemotherapy. It is well-known that menstruation alone does not accurately reflect ovarian function,²⁸ and biochemical measurement of a panel of established serum markers of ovarian function or reserve (including anti-Mullerian hormone (AMH), FSH, LH, estradiol, and inhibin-B) is the preferred clinical indicator. Monitoring of these hormone markers will help to detect chemotherapy-related ovarian toxicity prior to the physical sign of menstrual cessation. Measurement of AMH coupled with ultrasound-assisted antral follicle counting has been suggested as an effective means of assessing the ovarian reserve following breast

cancer chemotherapy.²⁹ Unfortunately, we were unable to include an analysis of the hormone markers to more accurately assess POF or ovarian reserve in the pooled patients from the RCTs in the current meta-analysis due to the lack of data.

Clinical studies have shown that age represents a significant risk factor for chemotherapy-induced POF.³⁰ Normal ovarian function itself is age-dependent, with ovarian function declining with age. Not surprisingly, older women experience a higher incidence of complete POF and permanent infertility following chemotherapy, as compared to their younger counterparts.^{31,32} Moreover, the risk of chemotherapy-induced amenorrhea is also directly related to the particular chemotherapy regimen.²⁶ Again, we were unable to perform subgroup analysis of POF based on age and chemotherapy regimen due to the limited data. The age range of the patients from the five RCTs was inconsistent, as were the chemotherapy regimens.

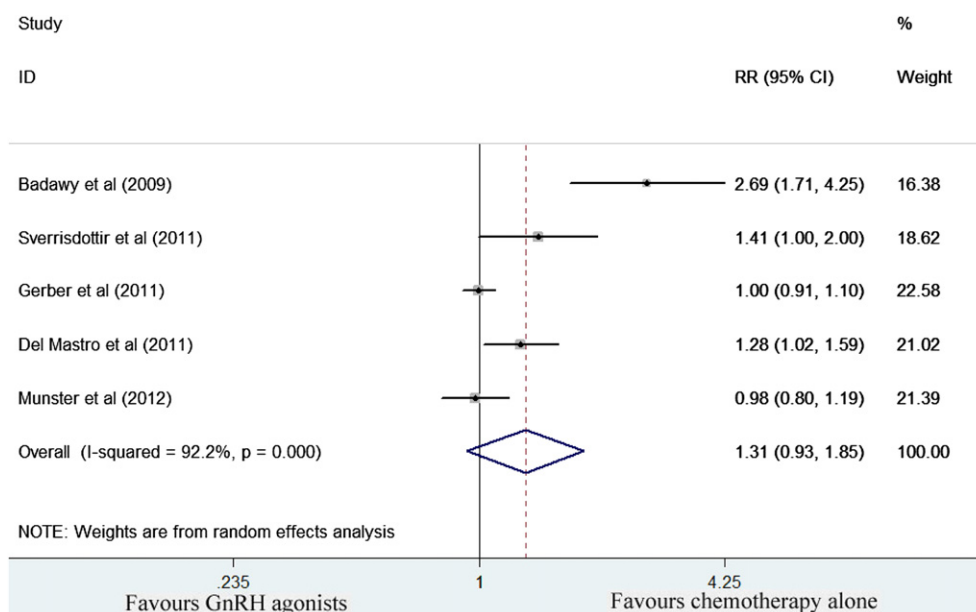


Fig. 3. Forest plots showing RR of resumption of menses rate of eligible studies comparing GnRH agonists plus chemotherapy with chemotherapy alone in a random effect model.

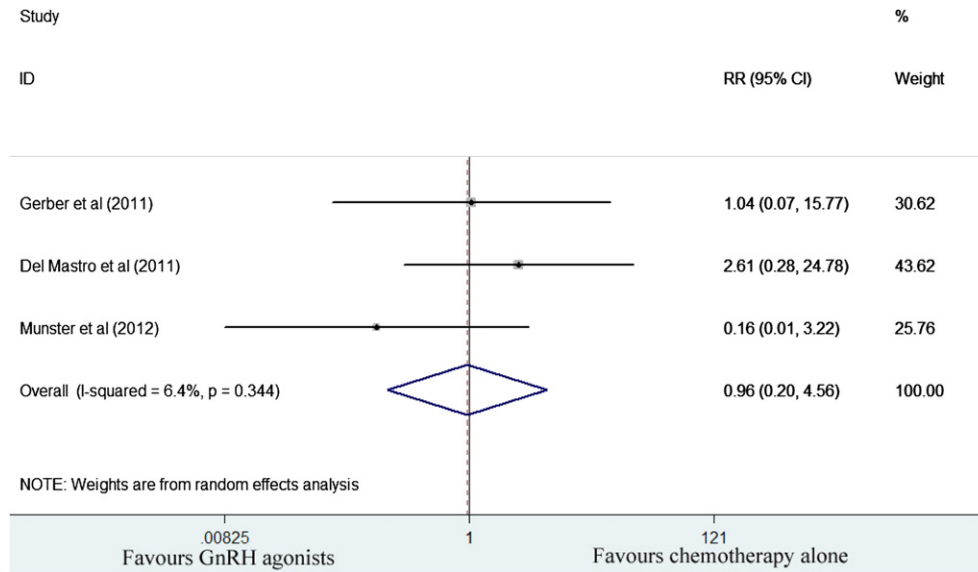


Fig. 4. Forest plots showing RR of spontaneous pregnancy rate of eligible studies comparing GnRH agonists plus chemotherapy with chemotherapy alone in a random effect model.

It should be noted that in older patients, the therapeutic effect of GnRH agonist may be decrease with decline in ovarian function.

Infertility is another potential complication of adjuvant chemotherapy that is experienced by some premenopausal women. These women may continue to menstruate or resume their menses after chemotherapy, despite having abnormal fertility. In the current meta-analysis, three of the RCTs^{12–14} evaluated spontaneous pregnancy rate but no statistically significant increase in spontaneous pregnancies was found for GnRH agonist treatment. However, the follow-up duration was too short to evaluate the real influence on fertility.

Despite the protective effect of GnRH agonist treatment on preserving ovarian function in premenopausal women treated with chemotherapy, a major concern of both patients and treating physicians is the potential side effects of the GnRH agonists themselves. GnRH agonists suppression of the reproductive axis causes typical symptoms of menopause, such as hot flashes, headaches, mood changes, sweating, and dry skin, as well as decreased bone density and possible predisposition to osteoporosis or bone fracture. However, the current meta-analysis found no statistically significant differences on adverse effects between the

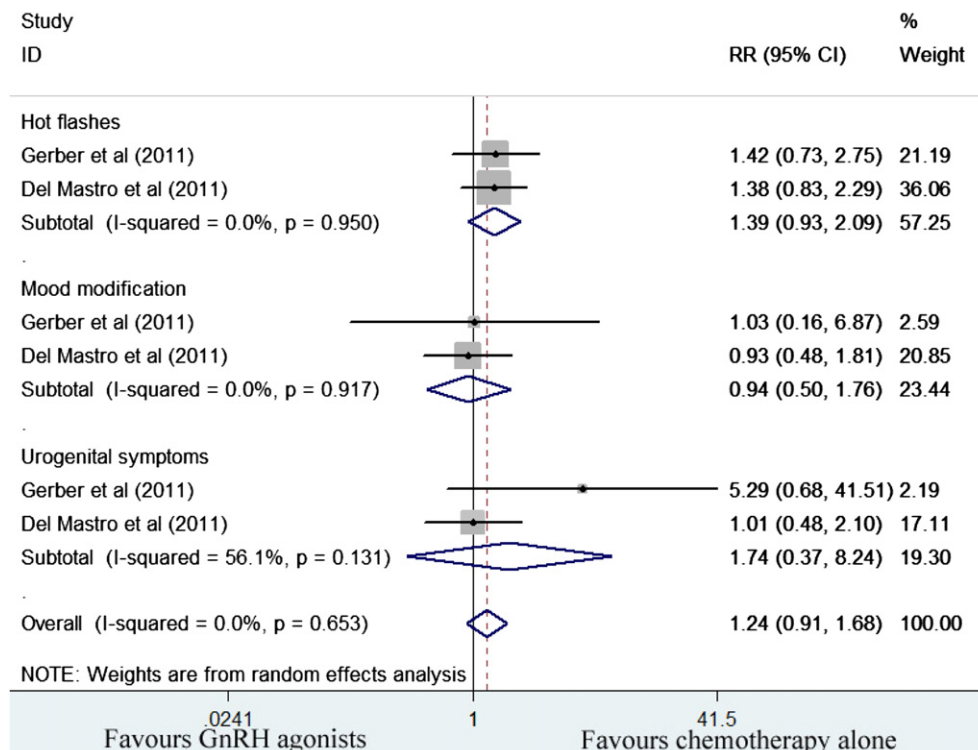


Fig. 5. Forest plots showing RR of adverse effects of eligible studies comparing GnRH agonists plus chemotherapy with chemotherapy alone in a random effect model.

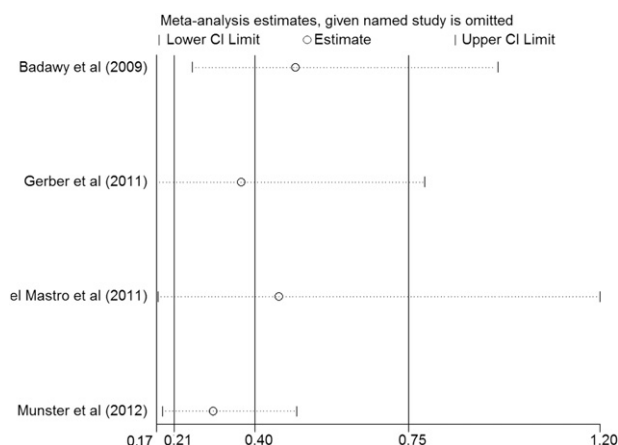


Fig. 6. RR and 95% CI by omitting each study from the included studies of POF rate.

women treated with GnRH agonists plus chemotherapy and those treated with chemotherapy alone ($p = 0.17$). Another concern is that GnRH agonists may reduce the effectiveness of the chemotherapy.³³ For this issue, a previous meta-analysis³⁴ assessed pooled data of 11,906 premenopausal women with early breast cancer from 16 RCTs and concluded that the administration of LH-RH agonists as adjuvant treatment did not reduce the effectiveness of the chemotherapy. Interestingly, most of the studies that did not meet the inclusion criteria for the current meta-analysis also found a beneficial effect of GnRH for reducing chemotherapy-associated POF in premenopausal patients with breast cancer.^{5,7–9} However, these studies lacked a control group, so that it was impossible to rule out age, rather than GnRH agonists treatment, as the main determinant of ovarian function preservation.

Although only RCTs were included in the current meta-analysis, several potential limitations exist that may have impacted the results. First, despite that fact that no language restrictions were applied to our literature search, only English languages RCTs were identified. It is possible that some relevant clinical data published in other languages may have been overlooked. Second, although there was no significant overall heterogeneity ($I^2 = 19.7\%$ in POF) among the five included studies, the obvious lack of uniform chemotherapy regimens, follow-up duration, and POF definition may have impacted the analysis. In addition, there is a major heterogeneity of the primary trials with regards to inclusion or exclusion of estrogen receptor positive patients and treatment with tamoxifen. Treatment with endocrine agents is a significant potential source of bias of individual studies. Third, women recruited to these five RCTs were between the ages of 13 and 47, and only one of the RCTs¹⁰ specifically concentrated on women under 40 years old. The inconsistent age of patients included in these RCTs may have lead to bias in the current meta-analysis. Fourth, environmental factors, such as occupational exposure to chemicals, has been implicated in POF²⁵; it is possible that an unidentified subgroup of patients represents a particular risk factor and biased the results of the current meta-analysis. Finally, the limited length of follow-up in the RCTs precludes our ability to determine the long-term impact of GnRH agonists on preservation of ovarian function and fertility.

Conclusion

Concurrent administration of GnRH agonists during chemotherapy appears to be an effective method to reduce the POF rate in premenopausal women with breast cancer. However, additional well-designed studies with larger and more diverse populations are

needed to confirm the protective effects of GnRH agonists against chemotherapy-induced ovarian damage and determine the related factors.

Conflict of interest statement

There are no conflicts of interest for any of the authors.

Acknowledgments

The authors wish to thank Shijun Wang for his assistance in preparing this manuscript.

References

- Jemal A, Siegel R, Xu J, Ward E. Cancer statistics, 2010. *CA Cancer J Clin* 2010;**60**: 277–300.
- Early Breast Cancer Trialists' Collaborative Group. Effects of chemotherapy and hormonal therapy for early breast cancer on recurrence and 15-year survival: an overview of the randomised trials. *Lancet* 2005;**365**:1687–717.
- Clarke M, Coates AS, Darby SC, Davies C, Gelber RD, Godwin J, et al. Adjuvant chemotherapy in oestrogen-receptor-poor breast cancer: patient-level meta-analysis of randomised trials. *Lancet* 2008;**371**:29–40.
- Tsutsui K, Bentley GE, Bedecarrats G, Osugi T, Ubuka T, Kriegsfeld LJ. Gonadotropin-inhibitory hormone (GnIH) and its control of central and peripheral reproductive function. *Front Neuroendocrinol* 2010;**31**:284–95.
- Recchia F, Sica G, De Filippis S, Saggio G, Rosselli M, Rea S. Goserelin as ovarian protection in the adjuvant treatment of premenopausal breast cancer: a phase II pilot study. *Anticancer Drugs* 2002;**13**:417–24.
- Gilani MM, Hasanazadeh M, Ghaemmaghami F, Ramazanzadeh F. Ovarian preservation with gonadotropin-releasing hormone analog during chemotherapy. *Asia-Pac J Clin Oncol* 2007;**3**:79–83.
- Recchia F, Saggio G, Amiconi G, Di Blasio A, Cesta A, Candeloro G, et al. Gonadotropin-releasing hormone analogues added to adjuvant chemotherapy protect ovarian function and improve clinical outcomes in young women with early breast carcinoma. *Cancer* 2006;**106**:514–23.
- Del Mastro L, Catzeddu T, Boni L, Bell C, Sertoli MR, Bighin C, et al. Prevention of chemotherapy-induced menopause by temporary ovarian suppression with goserelin in young, early breast cancer patients. *Ann Oncol* 2006;**17**:74–8.
- Urruticoechea A, Arnedos M, Walsh G, Dowsett M, Smith IE. Ovarian protection with goserelin during adjuvant chemotherapy for pre-menopausal women with early breast cancer (EBC). *Breast Cancer Res Treat* 2008;**110**:411–6.
- Badawy A, Elnashar A, El-Ashry M, Shahat M. Gonadotropin-releasing hormone agonists for prevention of chemotherapy-induced ovarian damage: prospective randomized study. *Fertil Steril* 2009;**91**:694–7.
- Sverrisdottir A, Nystedt M, Johansson H, Fornander T. Adjuvant goserelin and ovarian preservation in chemotherapy treated patients with early breast cancer: results from a randomized trial. *Breast Cancer Res Treat* 2009;**117**: 561–7.
- Gerber B, von Minckwitz G, Stehle H, Reimer T, Felberbaum R, Maass N, et al. Effect of luteinizing hormone-releasing hormone agonist on ovarian function after modern adjuvant breast cancer chemotherapy: the GBG 37 ZORO study. *J Clin Oncol* 2011;**29**:2334–41.
- Del Mastro L, Boni L, Michelotti A, Gamucci T, Olmeo N, Gori S, et al. Effect of the gonadotropin-releasing hormone analogue triptorelin on the occurrence of chemotherapy-induced early menopause in premenopausal women with breast cancer: a randomized trial. *J Am Med Assoc* 2011;**306**:269–76.
- Munster PN, Moore AP, Ismail-Khan R, Cox CE, Lacey M, Gross-King M, et al. Randomized trial using gonadotropin-releasing hormone agonist triptorelin for the preservation of ovarian function during (neo)adjuvant chemotherapy for breast cancer. *J Clin Oncol* 2012;**30**:533–8.
- Clowse ME, Behera MA, Anders CK, Copland S, Coffman CJ, Leppert PC, et al. Ovarian preservation by GnRH agonists during chemotherapy: a meta-analysis. *J Womens Health (Larchmt)* 2009;**18**:311–9.
- Chen H, Li J, Cui T, Hu L. Adjuvant gonadotropin-releasing hormone analogues for the prevention of chemotherapy induced premature ovarian failure in premenopausal women. *Cochrane Database Syst Rev* 2011. CD008018.
- Bedaiwy MA, Abou-Setta AM, Desai N, Hurd W, Starks D, El-Nashar SA, et al. Gonadotropin-releasing hormone analog cotreatment for preservation of ovarian function during gonadotoxic chemotherapy: a systematic review and meta-analysis. *Fertil Steril* 2011;**95**:906.e1–4–14.e1–4.
- Higgins JP, Green S, The Cochrane Collaboration. *Cochrane handbook for systematic reviews of interventions version 5.0.2* 2009.
- Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ* 2003;**327**:557–60.
- Begg CB, Mazumdar M. Operating characteristics of a rank correlation test for publication bias. *Biometrics* 1994;**50**:1088–101.
- Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;**315**:629–34.
- Conway GS. Premature ovarian failure. *Br Med Bull* 2000;**56**:643–9.

23. Rebar RW. Premature ovarian failure. *Obstet Gynecol* 2009;**113**:1355–63.
24. Christin-Maitre S, Braham R. General mechanisms of premature ovarian failure and clinical check-up. *Gynecol Obstet Fertil* 2008;**36**:857–61.
25. Beranger R, Hoffmann P, Christin-Maitre S, Bonnetterre V. Occupational exposures to chemicals as a possible etiology in premature ovarian failure: a critical analysis of the literature. *Reprod Toxicol* 2012;**33**:269–79.
26. Bines J, Oleske DM, Cobleigh MA. Ovarian function in premenopausal women treated with adjuvant chemotherapy for breast cancer. *J Clin Oncol* 1996;**14**:1718–29.
27. Tham YL, Sexton K, Weiss H, Elledge R, Friedman LC, Kramer R. The rates of chemotherapy-induced amenorrhea in patients treated with adjuvant doxorubicin and cyclophosphamide followed by a taxane. *Am J Clin Oncol* 2007;**30**:126–32.
28. Bukman A, Heineman MJ. Ovarian reserve testing and the use of prognostic models in patients with subfertility. *Hum Reprod Update* 2001;**7**:581–90.
29. Anderson RA, Themmen AP, Al-Qahtani A, Groome NP, Cameron DA. The effects of chemotherapy and long-term gonadotrophin suppression on the ovarian reserve in premenopausal women with breast cancer. *Hum Reprod* 2006;**21**:2583–92.
30. Lee SJ, Schover LR, Partridge AH, Patrizio P, Wallace WH, Hagerty K, et al. American Society of Clinical Oncology recommendations on fertility preservation in cancer patients. *J Clin Oncol* 2006;**24**:2917–31.
31. Moore HC. Fertility and the impact of systemic therapy on hormonal status following treatment for breast cancer. *Curr Oncol Rep* 2000;**2**:587–93.
32. Minton SE, Munster PN. Chemotherapy-induced amenorrhea and fertility in women undergoing adjuvant treatment for breast cancer. *Cancer Control* 2002;**9**:466–72.
33. Pritchard KI, Paterson AH, Paul NA, Zee B, Fine S, Pater J. Increased thromboembolic complications with concurrent tamoxifen and chemotherapy in a randomized trial of adjuvant therapy for women with breast cancer. National Cancer Institute of Canada Clinical Trials Group Breast Cancer Site Group. *J Clin Oncol* 1996;**14**:2731–7.
34. Cuzick J, Ambrosine L, Davidson N, Jakesz R, Kaufmann M, Regan M, et al. Use of luteinising-hormone-releasing hormone agonists as adjuvant treatment in premenopausal patients with hormone-receptor-positive breast cancer: a meta-analysis of individual patient data from randomised adjuvant trials. *Lancet* 2007;**369**:1711–23.